



Effect of surface roughness on pool boiling heat transfer at a heated surface having moderate wettability



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ABSTRACT

The effect of surface roughness on pool boiling heat transfer coefficient and critical heat flux (CHF) at a copper surface having moderate wettability was studied in saturated water. Copper surfaces were polished with sandpapers of different average surface roughness (R_a), ranging from 0.041 μm to 2.36 μm . Test measurements included static and dynamic contact angles for each of the nine surfaces tested. Although the surface roughness, R_a , moderately influenced the contact angles, pool boiling test results successfully correlated with the coefficient, C_{sf} , in the well-known Rohsenow correlation. The CHF showed noticeably strong dependence and an enhanced performance on the surface roughness as well. The CHF at the roughest surface ($R_a = 2.36 \mu\text{m}$) was 1625 kW/m^2 , which is approximately twice as much of that at the smoothest surface ($R_a = 0.041 \mu\text{m}$). The large increase in CHF with increasing surface roughness is considered to be a consequence of the capillary wicking from the surrounding liquid to the dry spot. A model for the CHF is obtained by modifying an existing correlation for pool boiling with the inclusion of the capillary wicking effect, and a comparison of the results with the experimental data shows good agreement when the wicking effect is included in the correlation.

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1. Introduction

Surface characteristics of heated surfaces are considered one of the most dominant factors affecting boiling heat transfer. Many researchers have studied the effect of surface roughness on the pool boiling heat transfer since Jakob's study of its effects on pool boiling [1]. Using rigorous experiments, Rohsenow [2] included the surface effects in his boiling correlation by matching the surface–fluid combination to the proper coefficient, C_{sf} . Subsequently, many studies have revealed the relationship between roughened surfaces and an increase in the number of nucleation sites, resulting in augmented boiling heat transfer [3–6]. Recent research efforts are focused on boiling heat transfer with a more complex surface exhibiting multiscale roughness [7], surface roughness accompanying a wettability change [8], and roughness in complex geometries [9]. A few attempts have been made to develop more accurate correlations between heat transfer coefficient h and roughness parameters [10–12]; these correlations were tabulated and compared with experimental data by Jones et al. [13]. Even though there are many correlations available, based on extensive experimentation with a variety of fluids over a range of pressures,

they do not all have a simple physical reasoning, unlike the Rohsenow correlation. Therefore, the first objective of the present study is to understand the nucleate boiling heat transfer enhancement due to surface roughness by establishing the relationship between the average roughness R_a and the coefficient C_{sf} in the conventional Rohsenow correlation.

In contrast to the clear relation between heat transfer coefficient and surface roughness [13], the effect of surface roughness on the critical heat flux (CHF) is still obscure. Many researchers believed that the CHF is triggered by the hydrodynamic instability [14] and does not include the effect of the surface condition, viz. surface roughness and wettability. The surface conditions were overlooked for a long time until Ramilison et al. [15] reported that they are important and concluded that the wettability effect on CHF is more prominent than the roughness effect. Many other studies related to the effect of surface roughness on the CHF were performed [16–18] and their data showed varying augmentation of CHF with increasing surface roughness. However, the extent of the augmentation was no more than $\pm 20\%$. Although surface roughness shows somewhat limited enhancement of CHF for typical metal surfaces, recent studies on the effect of surface roughness on hydrophilic silicon surfaces exhibited a much larger augmentation of CHF, unlike metal surfaces. Some researchers have focused on the combined effect of wettability and surface micro/nanostructures on silicon surfaces.

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Nomenclature

a	exponent in the Rohsenow correlation
A	constant
C	constant
C_p	specific heat [J/kg K]
C_{sf}	coefficient in the Rohsenow correlation
D_b	bubble diameter [m]
F	force [N]
g	gravity [m/s^2]
h	heat transfer coefficient [$\text{W/m}^2 \text{K}$]
h_{fg}	latent heat [J/kg]
H_b	bubble height [m]
m	exponent for surface roughness
n	exponent for heat flux
N	number of capillary tubes
P	pressure [Pa]
Pr	Prandtl number
q''	heat flux [W/m^2]
r	roughness factor
R	radius [m]
R_a	average roughness [m]
R_q	root mean squared roughness [m]
R_z	ten-point average roughness [m]
S	accommodation factor

S_m	mean spacing [m]
T	temperature [K]

Greek symbols

ΔT	wall superheat ($=T_w - T_{sat}$) [K]
θ	contact angle [$^\circ$]
λ_T	Taylor unstable wavelength [m]
μ	viscosity [N s/m^2]
ρ	density [kg/m^3]
σ	surface tension [N/m]

Subscripts

0	reference
C	capillary
CHF	critical heat flux
G	gravity
I	interface
l	liquid
M	momentum
sat	saturation
S	surface tension
v	vapor
w	wall

They have treated super-hydrophilic coating on well-designed artificial microstructures. These types of microstructures were fabricated on silicon wafer surfaces because micro-fabrication on silicon surfaces is fully developed as an accurate and mature technique. Some of these studies reported a surprisingly large increase in CHF on super-hydrophilic structured surfaces [7,19,20]. Rahman et al. [20] attained the CHF of as high as 2570 kW/m^2 on their hierarchically-structured silicon surface using the bio-template of the tobacco mosaic virus. The increase seems to be beyond the limit that can be attained by wettability alone. It is hypothesized that the capillary wicking effect prevents dry-out of the heated surface, thus resulting in augmented CHF. From the studies on silicon surfaces, it is optimistic to anticipate similar results on metal surfaces provided they are hydrophilic. O'Hanley et al. [8] pointed out that surface conditions including wettability, roughness, and porosity should be considered as separate parameters for precisely estimating the effect of each parameter on CHF. The present study is intended to start from roughened copper surfaces having moderate wettability, which means normal copper surfaces with contact angles of approximately $60\text{--}70^\circ$. Because it is not super-hydrophilic but still hydrophilic, it is expected that the wettability effect is relatively small, but capillary wicking due to the roughness change is still effective. Simultaneously, the static and dynamic contact angles are also measured in order to investigate the variation of contact angles with increasing surface roughness. On the basis of the experimental data, a CHF correlation including the capillary wicking is proposed that is appropriate for metal surfaces by using roughness parameter R_a .

2. Experimental setup

2.1. Surface preparation and characterization

Copper blocks of size $10 \text{ mm} \times 10 \text{ mm} \times 3 \text{ mm}$ were prepared to serve as test surfaces. The top and bottom surfaces were polished with a 600 grit sandpaper and cleaned by sonicating in

an isopropyl alcohol (IPA) solution, after which a $20\text{-}\Omega$ heating element was soldered to the bottom side of the copper block. In order to measure the temperature, a 30 AWG type T thermocouple was inserted into a hole drilled at the center of the copper block, where the hole is located at the 1.5 mm below the boiling surface. The top surfaces were polished again with various sandpapers having 2000, 600, 220, and 80 grit, and a compound that was a mixture of alumina and silicon oxide nanoparticles and tallow. The compound produced a smoother surface than that obtained by the sandpaper polishing process. The polishing was performed in one direction, which naturally made unidirectional scratches on the copper surfaces. The number of strokes for polishing was increased with increasing sandpaper grit because more strokes were required for making smoother surfaces. Although the number of strokes was 50 for the 220 and 80 grit sandpapers, it was 100 for the 600 grit sandpaper and approximately 300 for the 2000 grit sandpaper. A total of nine samples were prepared, two samples for each sandpaper and one for the compound.

The polished surfaces were characterized in two ways: wettability measurement and roughness measurement. Before the boiling test, the static and dynamic contact angles for each surface were measured using a goniometer (KRÜSS, DSA30). The volume of the sessile droplet was $11.5 \mu\text{l}$ for static contact angles. For advancing contact angles, the initial droplet of $30 \mu\text{l}$ was set up on the surface and the volume was increased to $50 \mu\text{l}$ by feeding water at a rate of $20 \mu\text{l/min}$. For receding contact angles, the droplet of $50 \mu\text{l}$ at the end of the advancing process shrank to zero volume at a rate of $20 \mu\text{l/min}$. After the boiling test, the static and dynamic contact angles were measured again to see how much the angles were changed during the test. The surface roughness was measured using the surface roughness tester (Mitutoyo, SJ-210). The tip radius of the probe was $2 \mu\text{m}$, the measuring force was 0.75 mN , and the evaluation length was 4 mm. Each sample was scanned at six different points to obtain the representative roughness. The roughness definition and the measurement technique follows standard ISO 4287-1997 practices.

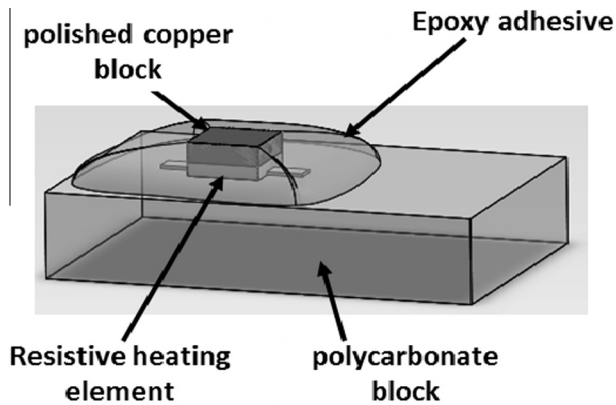


Fig. 1. Design of the test heater for the boiling experiment.

2.2. Pool boiling test section and test procedure

Fig. 1 shows the design of the test heater used in collecting data for the pool boiling experiments. A polished copper block bonded with the heating element is placed on the polycarbonate substrate and surrounded with epoxy adhesives (3M DP420) having low thermal conductivity of 0.18 W/m K , as shown in Fig. 1. The polished top surface is exposed to the exterior and acts as a boiling surface for the experiment. Fig. 2 shows a detailed drawing of the test chamber used in the experiment. A Graham-type coil condenser was used to condense and return the generated vapor to the chamber. The generated bubbles were observed through the sight glasses situated at the front and rear sides, and a high-speed camera was installed to capture them during the experiment. The saturated liquid temperature and vapor temperature were measured using type T thermocouple probes inserted inside the chamber. A pressure transducer ($0\text{--}172 \text{ kPa}$, $\pm 0.25\%$ accuracy) was connected to the chamber in order to measure the internal saturation pressure.

Approximately 2 l of distilled water was filled into the chamber with the test heater assembly placed at the center of the test chamber. Noncondensable gas in the water was removed by boiling the water with immersion heaters and band heaters for 1 h as a part of

the degassing process. The immersion heaters were then turned off, but the band heaters were kept on so that the water was maintained in its saturation state. Once the liquid temperature and internal pressure became stable, the test was initiated by applying the specified current to the test heater. The wall temperature, liquid temperature, vapor temperature, and pressure were measured, as well as the input current by measuring the voltage across a shunt resistor that was connected serially with the heater in the circuit. The input heat was calculated from the measured values of input current and voltage drop across the test heater. Once the wall temperature reached steady state, the current was increased to the next specified value. The temperatures were averaged every 20 s and the steady state was considered to exist when the temperature difference was less than 0.1°C between two consecutive average values. The power increment was varied to be $10\text{--}20 \text{ kW/m}^2$ at the incipience, 100 kW/m^2 at the nucleate boiling regime, and 50 kW/m^2 at near CHF. The boiling test was terminated when CHF occurred, defined by an abrupt increase in wall temperature by more than 20°C when compared to the value at the previous power input. The CHF was determined to be the average of the previous steady power input and the current power input which showed abrupt increase in wall temperature.

2.3. Uncertainty

The uncertainty analysis was performed by using the Kline and McClintock method for single-sample experiments [21]. The uncertainty of saturation temperature (T_{sat}) was calculated from the uncertainty of the pressure measurement. Because the pressure transducer has an accuracy of $\pm 0.25\%$ in full range ($0\text{--}172 \text{ kPa}$), the uncertainty of the saturation temperature is estimated as $\pm 0.1 \text{ K}$. The wall temperature was determined using the Fourier's law for conduction through the 1.5 mm copper block from the thermocouple location to the boiling surface. The wall temperature (T_w) has an uncertainty of $\pm 0.5 \text{ K}$. Input power, which is calculated from the measured voltage drop and current, has an error of $\pm 0.2 \text{ kW/m}^2$ at $q'' = 1000 \text{ kW/m}^2$. Heat loss through the epoxy insulation layer was estimated as 0.5% of total heat dissipation by a simple numerical calculation. The heater area's uncertainty was estimated as 5% of the total heated area. The uncertainty of

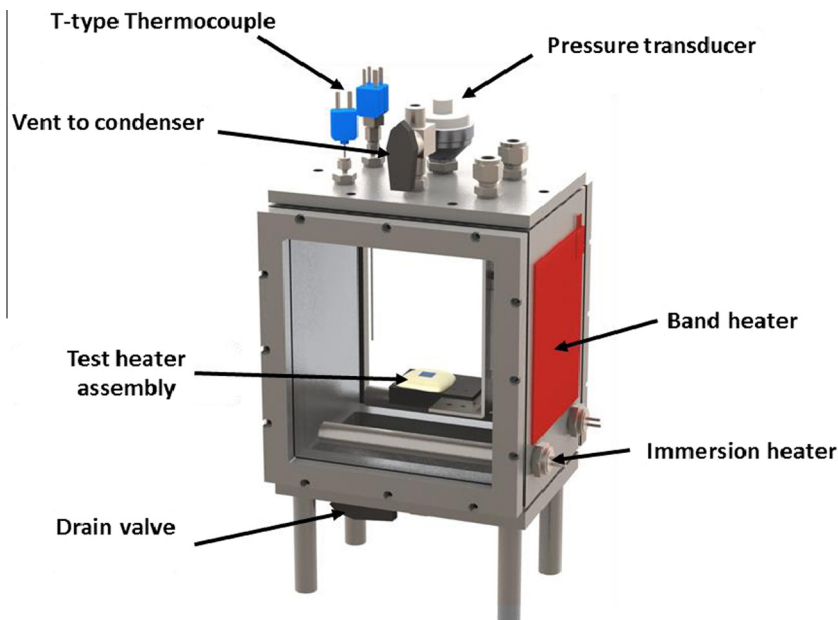


Fig. 2. Detailed drawing of the boiling test chamber.

the surface roughness was estimated statistically by assuming that the roughness data follow a normal distribution. With the confidence level at 95%, the average roughness of the second sample polished with 600 grit was $0.114 \pm 0.004 \mu\text{m}$ ($\pm 3.5\%$ error, smallest) and the roughness average of the first sample polished with 80 grit was $1.95 \pm 0.16 \mu\text{m}$ ($\pm 8.3\%$ error, largest).

3. Results and discussion

Table 1 shows the roughness parameters of all nine surfaces from the smoothest surface (compound) to the roughest surface (80 grit). The definition of the average roughness R_a and the root mean squared average R_q , which are commonly used roughness parameters, are as follows:

$$R_a = \frac{1}{n} \sum_{i=1}^n |y_i| \quad (1)$$

$$R_q = \sqrt{\frac{1}{n} \sum_{i=1}^n y_i^2} \quad (2)$$

R_z , as defined by the ISO standards, is the average of the five highest peaks and five deepest valleys. S_m refers to the mean spacing between peaks. S_m can provide information in the transverse direction, which is important for estimating the number of scratches per unit width. In some studies, a minimum height or a minimum depth is employed for ignoring small peaks, but in the present study no restriction was placed in counting the peaks. Because boiling heat transfer is related to the length scale of the heated surface, R_a was chosen as the representative roughness parameter in this study. The average roughness increased from $0.041 \mu\text{m}$ for the surface polished with compound to $2.36 \mu\text{m}$ for the surface polished with the 80 grit sandpaper. SEM images in Fig. 3 clearly show the different structures of the tested surfaces. The height (or the depth) of the unidirectional scratches increased with increasing R_a , whereas the number of scratches decreases from the smooth surface ($R_a = 0.057 \mu\text{m}$) to the rough surface ($R_a = 2.36 \mu\text{m}$). However, in case of the smoothest surface of $R_a = 0.041 \mu\text{m}$ (compound) the number of scratches also decreased, which means it had only sparse scratches. Mean spacing, S_m , expresses exactly the reverse trend as the number of scratches because S_m is inversely proportional to the number of scratches. It increases with increasing roughness from 2000 grit to 80 grit, as well as the surface polished with the compound.

Both the static and dynamic contact angles were measured for all nine surfaces. The unidirectional scratches require two different viewing angles for measuring the contact angles: parallel to the scratches and normal to the scratches. Fig. 4 depicts the sessile drops captured in the two directions for the representative surface with $R_a = 0.169 \mu\text{m}$ (600 grit #1). The contact angles captured parallel to the scratches normally have higher values because the

movement of contact lines is limited by the contact line pinning effect [22]. In contrast, the contact angles captured normal to the scratches showed lower values as the sessile drops spread easily through the scratches. Contact angles were measured before the boiling tests as well as after in order to investigate how much the contact angles changed during the tests. Static contact angles of all nine surfaces captured in the two directions are shown in Fig. 5. Static angles in the parallel direction, measured before the boiling tests, show only a slight increase with R_a , approximately 70° . The static contact angles in the normal direction are lower than those in the parallel direction because of the pinning and spreading effects, respectively, in those directions. It is noticeable that the static angle in the normal direction decreases with increasing roughness. This means that the spreading through the scratches increases as the surface becomes rougher. Because the spreading is caused by the capillary wicking, it can be inferred that the capillary wicking effect is increased with increasing surface roughness. Hence, the final averages of static contact angles in both directions have limited variation according to the surface roughness because of the opposing tendencies between contact line pinning in the parallel direction and spreading in the normal direction. The distribution of the averaged static contact angles was $64.5^\circ \pm 5^\circ$. The trends of the static contact angles after the boiling tests are the same as those before the boiling tests, but contact angles at both directions were increased by $5\text{--}15^\circ$ for all surfaces. The increase of the contact angles during the boiling tests is thought to result from the oxidation of the copper surfaces even at low temperatures near 100°C [23].

The dynamic contact angles (advancing and receding contact angles) were also measured as shown in Fig. 6, where the dynamic contact angles for the surface with $R_a = 0.169 \mu\text{m}$ (600 grit #1) are shown. Although the advancing contact angle is nearly constant throughout the growth period of the droplet, the receding contact angle exhibits a monotonic decrease during most of the shrinkage period of the droplet as a result of the contact line pinning effect. Because the receding of the triple contact line occurs only near the end of the shrinkage period, the receding contact angles were evaluated for only 5 s after the contact line actually moved. The advancing and receding contact angles with regard to R_a are shown in Fig. 7. The trend of the advancing contact angles with R_a is similar to that of the static contact angles except for somewhat higher values by approximately $10\text{--}15^\circ$. The receding contact angles also show a similar trend to the static contact angles, but they have much lower values of approximately 10° compared with the static contact angles. The variation of the receding contact angles according to R_a was not large, which implies negligible CHF variation with increasing R_a because the CHF is believed to be closely related to the receding contact angle [24]. This is why a large variation of CHF with R_a , which is observed later in the pool boiling curves, cannot be explained solely by the wettability effect.

Boiling curves for all the tested surfaces having different roughness are shown in Fig. 8. It is clearly demonstrated that the boiling curve shifts to the left as R_a increases, which indicates better nucleate boiling heat transfer performance with increasing R_a . This is consistent with previous studies that dealt with the effect of surface roughness on nucleate boiling heat transfer [4,6,25]. However, the change in CHF with increasing R_a in Fig. 8 is considerably large. The CHF was merely 775 kW/m^2 at the smoothest surface of $R_a = 0.041 \mu\text{m}$, whereas CHF at the roughest surface of $R_a = 2.36 \mu\text{m}$ was as much as 1625 kW/m^2 , which is more than the double of that at the smoothest surface. Therefore, the effects of surface roughness on nucleate boiling heat transfer and CHF must be investigated more thoroughly.

First, nucleate boiling heat transfer is known to be strongly affected by surface roughness as indicated in various studies. The surface roughness is considered to enhance the nucleate boiling

Table 1
Roughness parameters for all nine surfaces tested.

Sample	Roughness [μm]			
	R_a	R_q	R_z	S_m
Compound	0.041	0.055	0.438	20.9
2000 grit #1	0.061	0.080	0.699	15.1
2000 grit #2	0.057	0.076	0.699	13.1
600 grit #1	0.169	0.231	1.93	16.3
600 grit #2	0.114	0.157	1.41	14.7
220 grit #1	0.590	0.787	5.47	24.8
220 grit #2	0.997	1.31	8.57	27.2
80 grit #1	1.95	2.49	13.8	40.7
80 grit #2	2.36	3.01	16.3	44.5

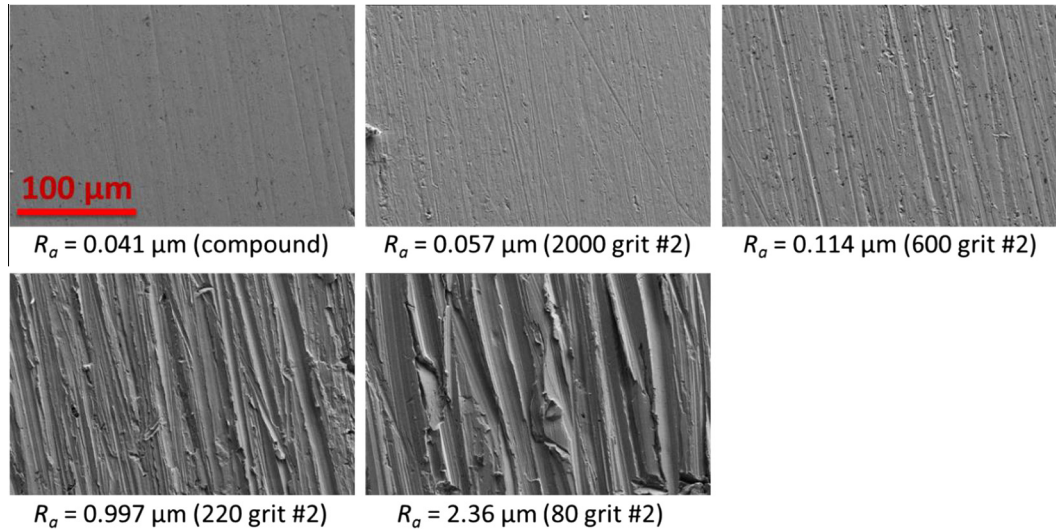


Fig. 3. SEM images of surfaces having different roughness ($\times 1000$).

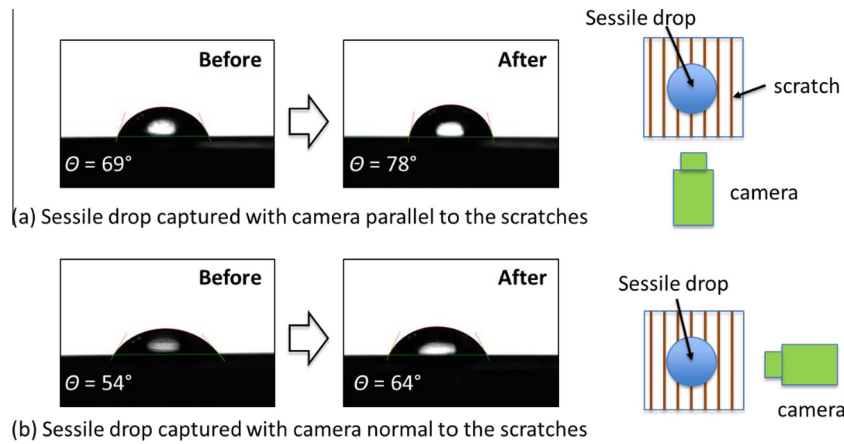


Fig. 4. Static contact angles at two directions: (a) parallel to the scratches, (b) normal to the scratches ($R_a = 0.169 \mu\text{m}$, 600 grit #1).

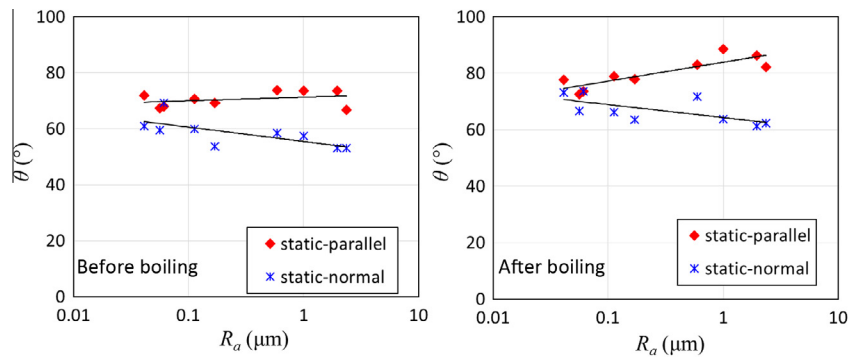


Fig. 5. Static contact angles for all nine surfaces at the parallel and normal directions.

because of the increased number of active nucleation sites. In fact, the coefficient, C_{sf} , in the well-known Rohsenow correlation for nucleate boiling heat transfer should be a function of surface roughness and wettability. Some researchers have suggested empirical correlations for the roughness effect on nucleate boiling heat transfer [10–12]. However, it is necessary to incorporate the effect of surface roughness directly into C_{sf} because the constant should be affected by both contact angle and roughness. In the

present study, all polished copper surfaces exhibit moderate contact angles of approximately 65° in the parallel and normal directions. Because the contact angles are maintained to be within the marginal range for all nine cases, C_{sf} is believed to be correlated according to the surface roughness.

$$\frac{q''}{\mu_l h_{fg}} \left[\frac{\sigma}{g(\rho_l - \rho_v)} \right]^{1/2} = \left(\frac{1}{C_{sf}} \right)^{1/a} Pr_l^{-1/a} \left[\frac{C_{pl} \Delta T}{h_{fg}} \right]^{1/a} \quad (3)$$

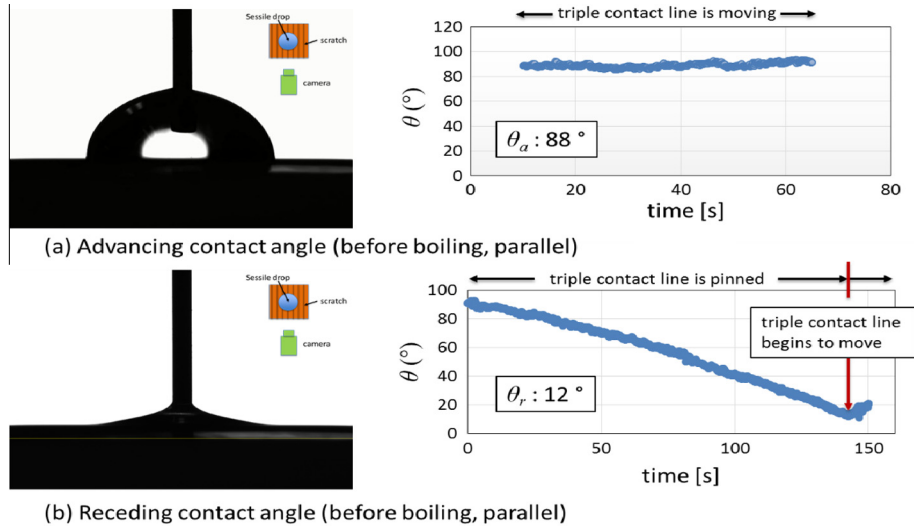


Fig. 6. Measurement of dynamic contact angles ($R_a = 0.169 \mu\text{m}$): (a) advancing contact angle (before boiling, parallel), (b) receding contact angle (before boiling, parallel).

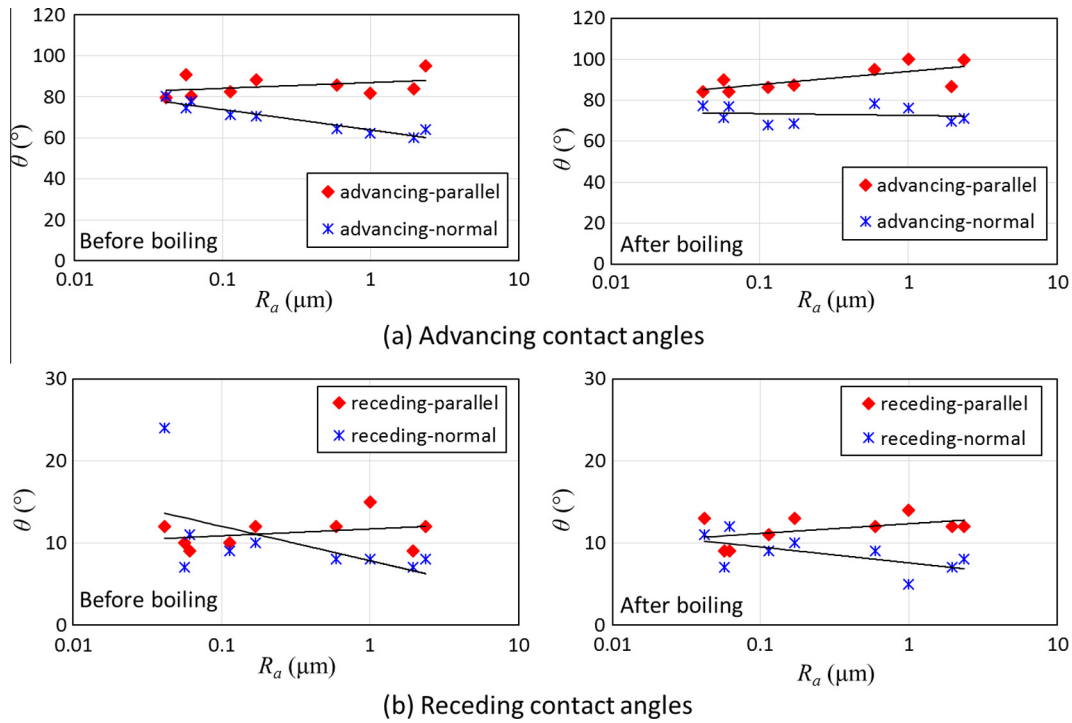


Fig. 7. Advancing and receding contact angles with R_a for two different directions.

Eq. (3) is the Rohsenow correlation for water, where a is reported as 0.33 [2]. Converting Eq. (3) into a dimensionless form of heat flux (q'') and heat transfer coefficient (h), the equation yields:

$$\frac{h}{h_0} = \left(\frac{1}{C_{sf}} \right) \left(\frac{q''}{q''_0} \right)^n \quad (4)$$

$$\text{where, } h_0 = \frac{\mu_l C_{pl}}{Pr_l} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2}, \quad q''_0 = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2},$$

$n = 1 - a$.

The coefficient C_{sf} can be regarded as a function of surface roughness R_a only, because the contact angles are similar; it is assumed as

$$\frac{1}{C_{sf}} = f(R_a) = A \left(\frac{R_a}{R_{a0}} \right)^m \quad (5)$$

R_{a0} is $0.4 \mu\text{m}$, which is a representative surface roughness for typical metal surfaces [11]. Exponent m and coefficient A are assumed constants. Combining Eqs. (4) and (5), the final form results in

$$\frac{h}{h_0} = A \left(\frac{R_a}{R_{a0}} \right)^m \left(\frac{q''}{q''_0} \right)^n \quad (6)$$

The value of A in Eq. (6) can be computed from the experimental data by assuming the value for n as 0.67 from the original Rohsenow correlation. It is also possible to study the effect of roughness on the variation in parameter n by performing the best fit from the experimental data for h and q . Fig. 9 shows the relation between dimensionless h and dimensionless q'' for all the surfaces tested

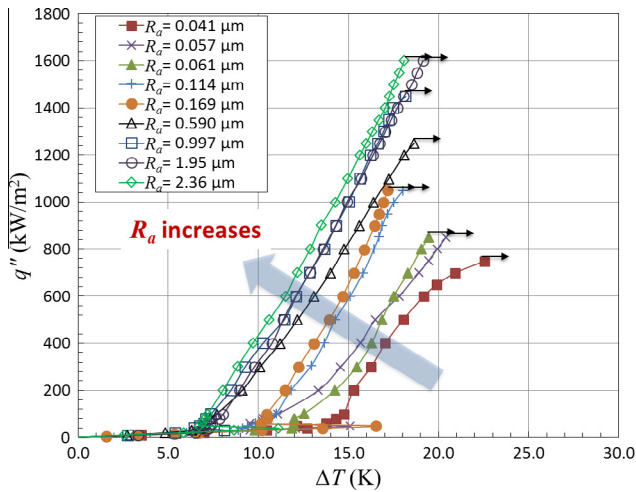


Fig. 8. Boiling curves with increasing R_a from 0.041 μm to 2.36 μm .

in this study. Exponent n of the trend line for each R_a is determined and plotted against as shown in the inset, indicating a variation in n according to dimensionless R_a . Exponent n is found to decrease moderately from 0.80 to 0.63 with increasing R_a , as shown by

$$n = 0.698 \left(\frac{R_a}{R_{a0}} \right)^{-0.057} \quad (7)$$

By substituting n values and all the measured data of h , q'' , and R_a for Eq. (6), constants A and m are obtained from the best fitting for the data. The final correlation of Eq. (6) assuming $n = 0.67$ as a constant is

$$\frac{h}{h_0} = 84.3 \left(\frac{R_a}{R_{a0}} \right)^{0.109} \left(\frac{q''}{q''_0} \right)^{0.67} \quad (8)$$

and the final correlation for n as a function of R_a in Eq. (7) is

$$\frac{h}{h_0} = 83.3 \left(\frac{R_a}{R_{a0}} \right)^{0.136} \left(\frac{q''}{q''_0} \right)^{0.698(R_a/R_{a0})^{-0.057}} \quad (9)$$

Eqs. (8) and (9) can be compared with the experimental data. Fig. 10 provides a comparison of the experimental data with Eq. (8) assuming $n = 0.67$. Most of the dimensionless heat transfer coefficients obtained from the correlation (Eq. (8)) are in good

agreement with the experimental data except for the high heat flux region on the rough surface. Because the nucleate boiling heat transfer is more affected by partial blanketing of vapor approaching CHF, the correlation for the nucleate boiling seems to deviate at a heat flux exceeding 80% of the CHF values. Nevertheless, the mean absolute error was calculated to be 6.7%, which is satisfactory. The correlation using more complicated n as a function of R_a (Eq. (9)) was also compared with the current data, and it showed an absolute error of 5.3%. Though the correlation using a more accurate estimation of n gives slightly better accuracy, it is recommended to use a simpler form with $n = 0.67$. In summary, the coefficient C_{sf} is proposed as the following correlation with dimensionless R_a :

$$\frac{1}{C_{sf}} = 84.3 \left(\frac{R_a}{R_{a0}} \right)^{0.109} \quad (10)$$

Although C_{sf} is a function of contact angle as well as surface roughness, Eq. (10) is likely to be applicable to surfaces having moderate wettability similar to the surfaces tested in the present study. A more generalized and complete correlation of C_{sf} should be acquired through further studies on the roughness effect for various wetting surfaces.

The boiling curves in Fig. 8 exhibit significant changes of the CHF with surface roughness. Many researchers have simply reported this behavior as augmentation of CHF due to surface roughness. However, the extent of augmentation demonstrated by the experimental data shows large differences according to the surface type, such as metal surfaces or silicon wafer surfaces. CHF variation with changing roughness on the metal surfaces has been reported to be approximately 10–20% in general [15,18]. In contrast, recent studies on CHF enhancement with silicon wafer surfaces show much larger augmentations with increasing roughness factor r , where roughness factor r is defined as the ratio of the actual contact area to the projected area [7,19,26,27].

Table 2 summarizes various studies regarding the effect of surface roughness on CHF for both metal surfaces and silicon wafer surfaces. CHF enhancement for metal surfaces is at most 10–20% except for the data measured by Ferjančič and Golobič [28] for etched surfaces, which exhibited approximately a 50% enhancement. Surfaces polished with sandpapers normally showed a 10–20% increase in CHF, whereas surfaces etched in diluted acid exhibited larger enhancements, as much as 51%. In contrast, the

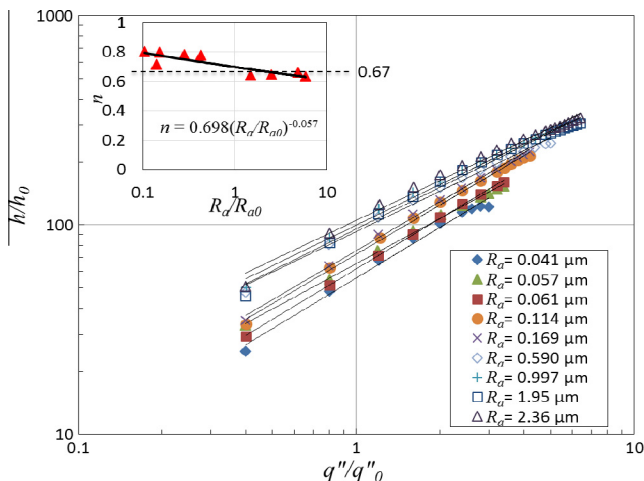


Fig. 9. Relation between dimensionless h and dimensionless q'' according to the surface roughness.

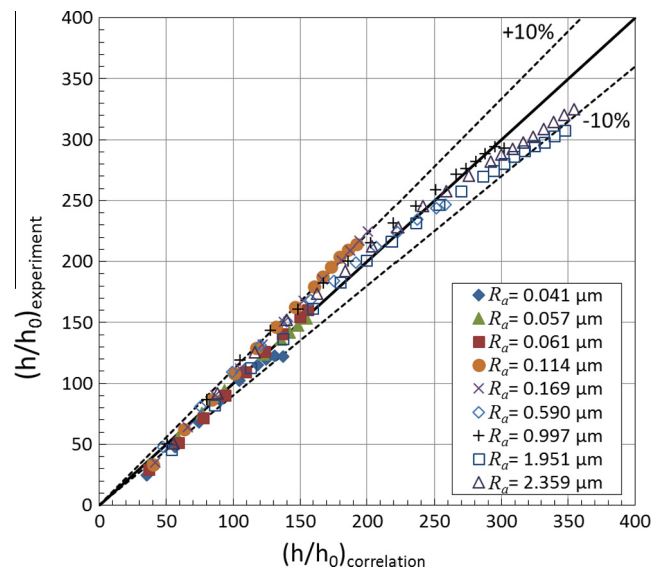


Fig. 10. Comparison between experimental data and the correlation (Eq. (8)).

Table 2

Previous studies regarding the effect of surface roughness on CHF for metal and silicon surfaces.

Surface type	Author	Surface	Fluid	Roughness parameter	Roughness range	Wettability	θ	Effect on CHF	Correlation
Metal	Ramilison et al. [15] [*]	Copper	Various	Rms roughness, R_q	2–12 μm	Various	0–40°	±10–20% enhancement	$\frac{q''_{CHF}}{q''_{CHF,z}} = 0.0336(\pi - \theta_r)^{3.0}(R_q)^{0.125}$
	Ferjančič and Golobič [16,28]	SS 302, steel 1010	FC-72, water	R_a	0.02–1.5 μm	Not reported	–	Max 51% increase (etched surface)	$q''_{CHF} = a \ln(R_a) + b$
	Ahmad et al. [17]	Copper	R-123	P_a (R_a without cut-off filter)	3.5 μm	Highly wetting fluid	–	225 kW/m ²	No correlation
	Guan et al. [18]**	Brass	Pentane, hexane, FC-72	R_q	0.15–5 μm	Highly wetting fluids	–	Increases 1.15 times	$\frac{q''_{CHF}}{q''_{CHF,smooth}} = \sqrt{\frac{A^2 + \sqrt{A^4 + 4}}{2}}$
Si wafer	Honda et al. [26]	Si	FC-72	r	2.2	Not reported	–	Increases 1.8–2.3 times	No correlation
	Chu et al. [7]***	Si	Water	r	3.6–13.3	Super-hydrophilic (micro/nanostructure)	0°	Max. 2500 kW/m ²	$K = \frac{(1+\cos\theta)}{16} \left[\frac{2(1+r\cos\theta_{gc})}{\pi(1+\cos\theta)} + \frac{\pi}{4}(1+\cos\theta)\cos\phi \right]^{1/2}$
	O'Hanley et al. [8]	Si	Water	$R_a/R_q/R_z/r$	0.4–7.2 μm (R_a), 0–1.44 (r)	Hydrophilic and hydrophobic	0–140°	Roughness has no effect	No correlation
	Kim et al. [27]	Si	Water	r	6.46–10.75	Super-hydrophilic (nanostructure)	6.97–10.5°	1145–2143 kW/m ²	$q''_{CHF} = q''_{CHF,Kandlikar} + Bh_{fg}\rho_l(1-\alpha)W^2/\lambda_{RT}^2$
	Guan et al. [18]**	Si	Pentane, hexane, FC-72	d (hoodoo dia.)	10–80 μm	Highly wetting fluids	–	Increases 1.05–1.67 times	$\frac{q''_{CHF}}{q''_{CHF,smooth}} = \sqrt{\frac{A^2 + \sqrt{A^4 + 4}}{2}}$

^{*} θ_r : receding contact angle.^{**} A : a dimensionless variable including the fluid properties and the wicking velocity.^{***} K : CHF parameter coming from the relation $q''_{CHF} = K \times h_{fg}\rho_l^{1/2}[\sigma g(\rho_l - \rho_v)]^{1/4}$.

enhancement of CHF on silicon wafer surfaces is more pronounced and shows more than a 100% increase in CHF, depending on the roughness factor, r . The considerably higher enhancement of CHF on the silicon surfaces can be attributed to the increased capillary wicking phenomenon with increasing surface roughness. Moreover, because most silicon surfaces were super-hydrophilic, aided by the nanostructures, it is believed that capillary wicking is more intensive in the microgeometries on the silicon surfaces. Because capillary pressure exists if the surface is hydrophilic, the copper surfaces having moderate wettability in the present study certainly have the capillary wicking effect.

It is already discussed that the static and dynamic contact angles viewed in the normal-to-the-scratch direction consistently decrease with increasing R_a because of the capillary wicking (Figs. 5 and 7). Fig. 11 shows the model of a hovering bubble on the modified surface structures created by the current surface roughness following the original model proposed by Haramura and Katto [29]. They explained that CHF occurs at the dry-out when the liquid sublayer beneath the hovering bubble completely evaporates. In the case of a perfectly smooth surface, the liquid amount in the liquid sublayer is established at the initial stage of the bubble cycle without any additional supply. However, for roughened surfaces, adjacent liquid is likely to be transported to the sublayer by capillary wicking. This additional liquid supply could delay the dry-out of the liquid sublayer at the same q'' where CHF occurs at the perfectly smooth surface, thus augmenting the CHF at the roughened surface. Because the capillary wicking increases further with surface roughness, the augmentation of CHF is predicted to increase as well. The CHF measurements in the present study clearly demonstrate this fact with CHF increasing from 775 kW/m² at the smoothest surface ($R_a = 0.041 \mu\text{m}$) to 1625 kW/m² at the roughest surface ($R_a = 2.36 \mu\text{m}$).

The CHF model, including the wettability effect on the conventional hydrodynamic model, was first proposed by Kandlikar [24]. He successfully added the wettability effect in his CHF model by balancing momentum change due to evaporation with the gravity

and surface tension forces exerted on the bubble. Recently, some researchers considered the contribution of surface roughness to CHF as an additional term to Kandlikar's CHF model. Ahn et al. [19] ascribed the large increase of CHF on their super-hydrophilic surfaces to the existence of additional liquid spreading above the wettability effect. This liquid spreading is thought to be another expression of capillary wicking. As shown in Table 2, Chu et al. [7] modified the Kandlikar correlation to accommodate the effect of surface roughness by changing $\cos\theta$ to $r\cos\theta$, where r is the roughness factor. Kim et al. [27] also proposed a CHF correlation including the surface roughness effect by adding a capillary wicking term to the Kandlikar model. Although various attempts were made to fit the capillary effect into the CHF model for explaining the large augmentation on silicon surfaces, an appropriate CHF correlation accounting for the roughness effect on metal surfaces is difficult to analyze. Because a different roughness parameter r is normally used for silicon surfaces, a new correlation using R_a is required to account for the roughness effect on CHF for metal surfaces.

For CHF modeling on the present metal surfaces, it can be thought of as having many small capillary tubes beneath the vapor bubble as shown in Fig. 12. Force balance between momentum change due to evaporation and forces acting on the vapor bubble can then be analyzed in a similar way to Kandlikar's original approach, with an additional capillary force term. Quan et al. [30] first considered capillary wicking due to surface roughness as an additional force term in the force balance equation:

$$F_M = F_{S,1} + F_{S,2} + F_G + F_C \quad (11)$$

where F_M is the momentum change due to the evaporation at the interface and $F_{S,1}$, $F_{S,2}$, F_G , and F_C refer to surface force at the bottom of the bubble, surface force at the top of the bubble, gravity force, and capillary wicking force, respectively.

$$F_M = \frac{1}{\rho_v} \left(\frac{q''_l}{h_{fg}} \right)^2 H_b \quad (12)$$

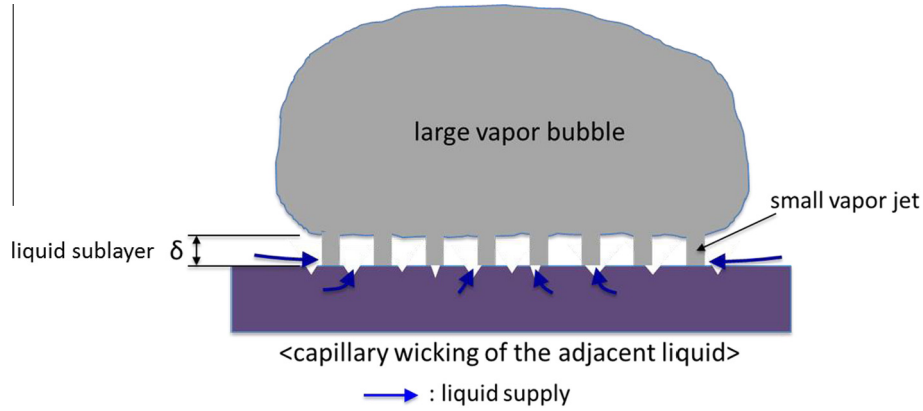


Fig. 11. Structure of a hovering bubble including liquid supply by capillary wicking (modification of the original drawing by Haramura and Katto [29]).

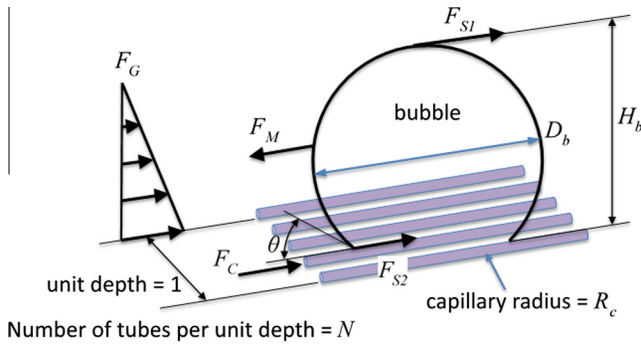


Fig. 12. Small capillary tubes beneath the bubble for modeling capillary wicking due to surface roughness.

$$F_C = N\pi R_c^2 P_c = \pi N R_c^2 \frac{2\sigma \cos \theta}{R_c} = 2\pi N R_c \sigma \cos \theta \quad (20)$$

The number of capillary tubes N is also considered to be inversely proportional to the mean spacing of the surface roughness S_m :

$$N \propto \frac{1}{S_m} \quad (21)$$

Using Eqs. (19) and (22), we can arrive at a relationship for capillary force as

$$F_C = 2\pi C \left(\frac{R_a}{S_m} \right) \sigma \cos \theta \quad (22)$$

In Eq. (22), C is a constant of proportionality.

Substituting all force terms, Eqs. (12)–(15) and (22), into Eq. (11) results in

$$\frac{1}{\rho_v} \left(\frac{q''_l}{h_{fg}} \right)^2 H_b = \sigma + \sigma \cos \theta + \frac{1}{2} g (\rho_l - \rho_v) H_b^2 + 2\pi \sigma C \left(\frac{R_a}{S_m} \right) \cos \theta \quad (23)$$

Rearranging Eq. (23) obtains

$$q''_l = h_{fg} \rho_v^{1/2} [\sigma g (\rho_l - \rho_v)]^{1/4} \left[\frac{2}{\pi} + \frac{\pi}{4} (1 + \cos \theta) + \frac{4C \cos \theta}{1 + \cos \theta} \left(\frac{R_a}{S_m} \right) \right]^{1/2} \quad (24)$$

If the surface roughness is neglected, then the last term in the bracket on the right side in Eq. (24) disappears and the remaining equation is the same as that of Kandlikar's CHF model of the horizontally oriented surface. Once heat flux at the interface q''_l is derived, CHF at the heated wall can be obtained through the following relation between q''_l and q''_{CHF} [24]:

$$q''_{CHF} = \frac{(1 + \cos \theta)}{16} q''_l \quad (25)$$

Though Eq. (25) is based on physical reasoning, the predicted value on the mirror surface ($R_a \sim 0$) exhibits a large difference from the empirical data reported by some researchers. Although Eq. (25) gives a CHF value of 1205 kW/m² for a mirror surface having a contact angle of 50°, the measured CHF ranges from 700 kW/m² to 1000 kW/m² [7,19,27]. To accommodate the difference in CHF at the mirror surface, a factor S is introduced for the final CHF correlation:

$$q''_{CHF} = S \times h_{fg} \rho_v^{1/2} [\sigma g (\rho_l - \rho_v)]^{1/4} \times \frac{(1 + \cos \theta)}{16} \left[\frac{2}{\pi} + \frac{\pi}{4} (1 + \cos \theta) + \frac{4C \cos \theta}{1 + \cos \theta} \left(\frac{R_a}{S_m} \right) \right]^{1/2} \quad (26)$$

$$F_{S,1} = \sigma \quad (13)$$

$$F_{S,2} = \sigma \cos \theta \quad (14)$$

$$F_G = \frac{1}{2} g (\rho_l - \rho_v) H_b^2 \quad (15)$$

where the gravity force is formulated with regard to a horizontal heated surface facing upward. Bubble height H_b is related to bubble diameter D_b by geometry and D_b is assumed to be half of the Taylor unstable wavelength λ_T in the same way that the Kandlikar model does.

$$H_b = \frac{D_b}{2} (1 + \cos \theta) \quad (16)$$

$$D_b = \frac{\lambda_T}{2} = \pi \left[\frac{\sigma}{g (\rho_l - \rho_v)} \right]^{1/2} \quad (17)$$

Capillary force can be derived starting from the Young–Laplace equation for capillary pressure inside a small tube:

$$P_c = \frac{2\sigma \cos \theta}{R_c} \quad (18)$$

where R_c refers to the radius of the capillary tube. Now, it is assumed that many capillary tubes lie beneath a large vapor bubble. Then, R_c is reasonably inferred to be proportional to the average roughness, R_a :

$$R_c \propto R_a \quad (19)$$

The capillary force is determined by multiplying the capillary pressure and the tube cross-sectional area:

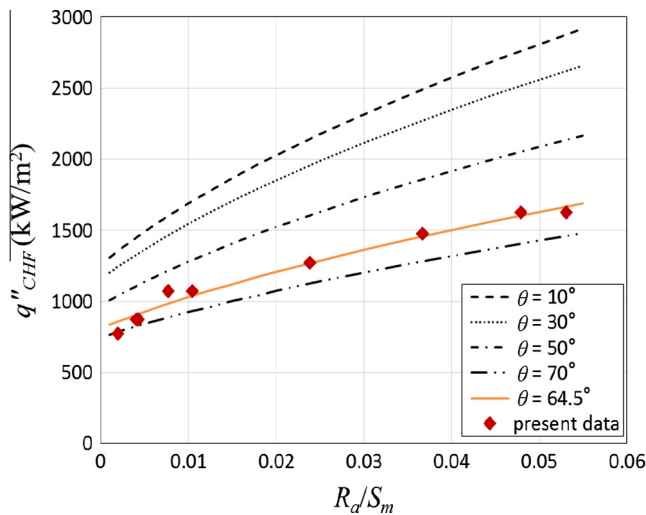


Fig. 13. Trend of Eq. (26) at different contact angles of 10°, 30°, 50°, and 70° and comparison with the measured data at the contact angle of 64.5°.

In the present study, the accommodation factor S and the constant C were obtained from Eq. (26) using the measured data—contact angles, roughness parameters (R_a and S_m), and CHF. The static contact angles captured in the parallel and normal directions to the scratches were averaged at all roughened surfaces. The averaged static contact angle, which was calculated to be 64.5°, is employed for the calculation of S and C because the average at each surface showed similar values within $\pm 5^\circ$. The S and C values are found to be 0.811 and 87.8 for the current data set. The current value of accommodation factor S , 0.811, is almost identical to the value of 0.823 suggested by Ahn et al. [19]. As a result, the curves calculated from the final CHF correlation (Eq. (26)) at different contact angles (10°, 30°, 50°, and 70°) exhibited a trend with increasing surface roughness at different wettability in Fig. 13. The present data are also plotted in the figure with the relevant curve obtained at 64.5°, the average static contact angle measured in the present study. The data show good agreement with the correlation, with a mean absolute error of 4.1%. Fig. 13 also indicates the possibility of attaining higher CHF than 2500 kW/m² at the super-hydrophilic roughened surfaces ($\theta = 10^\circ$ and 30° in the figure). Additional studies on super-hydrophilic and hydrophobic metal surfaces would give us full understandings of the combined effect of roughness and wettability on CHF in boiling heat transfer.

4. Conclusions

The effect of surface roughness on the nucleate boiling heat transfer and CHF at copper surfaces having moderate contact angles in pool boiling of saturated water was investigated. The average roughness R_a was varied from 0.041 μm to 2.36 μm , and static and dynamic contact angles were measured before and after each boiling test for all nine surfaces. From the test results and discussion, the following conclusions are deduced.

- Both static and dynamic contact angles for tested surfaces showed similar behavior with increasing surface roughness. The contact angles viewed in the parallel direction tend to increase with R_a , while the contact angles viewed in the normal direction tend to decrease with R_a . This behavior is considered to result from the contact line pinning (parallel direction) and capillary wicking (normal direction).

- It was confirmed that surface roughness enhances nucleate boiling heat transfer because of the increased number of active nucleation sites. For a boiling surface having moderate wettability, coefficient C_{sf} in the Rohsenow correlation was successfully correlated with regard to the dimensionless average roughness. The heat transfer coefficient calculated from the proposed correlation shows good agreement with the experimental data.
- CHF exhibited considerable augmentation with increasing surface roughness. It increased from 775 kW/m² at the smoothest surface ($R_a = 0.041 \mu\text{m}$) to 1625 kW/m² at the roughest surface ($R_a = 2.36 \mu\text{m}$). The large augmentation is believed to result from capillary wicking on the roughened surface. An appropriate CHF correlation including the capillary wicking effect was proposed for metal surfaces and it showed good agreement with the experimental data (average 4% difference).

References

- [1] M. Jakob, *Heat Transfer*, vol. 1, Wiley, New York, 1949, pp. 631–642.
- [2] W.M. Rohsenow, *A Method of Correlating Heat Transfer Data for Surface Boiling of Liquids*, MIT Division of Industrial Cooperation, Cambridge, Mass, 1951.
- [3] H.M. Kurihara, J.E. Myers, The effects of superheat and surface roughness on boiling coefficients, *AIChE J.* 6 (1) (1960) 83–91.
- [4] R.I. Vachon, G.E. Tanger, D.L. Davis, G.H. Nix, Pool boiling on polished and chemically etched stainless-steel surfaces, *J. Heat Transfer* 90 (2) (1968) 231–238.
- [5] B.B. Mikic, W.M. Rohsenow, A new correlation of pool-boiling data including the effect of heating surface characteristics, *J. Heat Transfer* 91 (2) (1969) 245–250.
- [6] A. Luke, Pool boiling heat transfer from horizontal tubes with different surface roughness, *Int. J. Refrig* 20 (8) (1997) 561–574.
- [7] K.H. Chu, Y.S. Jung, R. Enright, C.R. Buie, E.N. Wang, Hierarchically structured surfaces for boiling critical heat flux enhancement, *Appl. Phys. Lett.* 102 (15) (2013) 151602.
- [8] H. O'Hanley, C. Coyle, J. Buongiorno, T. McKrell, L.W. Hu, M. Rubner, R. Cohen, Separate effects of surface roughness, wettability, and porosity on the boiling critical heat flux, *Appl. Phys. Lett.* 103 (2) (2013) 024102.
- [9] A.S. Moita, E. Teodori, A.L.N. Moreira, Influence of surface topography in the boiling mechanisms, *Int. J. Heat Fluid Flow* 52 (2015) 50–63.
- [10] M.G. Cooper, Saturation nucleate pool boiling—a simple correlation, *Int. Chem. Eng. Symp. Ser.* 86 (2) (1984) 785–793.
- [11] D. Gorenflo, D. Kenning, *H2 Pool Boiling*, Springer, Berlin Heidelberg, 2010, pp. 757–792.
- [12] W. Leiner, Heat transfer by nucleate pool boiling—general correlation based on thermodynamic similarity, *Int. J. Heat Mass Transfer* 37 (5) (1994) 763–769.
- [13] B.J. Jones, J.P. McHale, S.V. Garimella, The influence of surface roughness on nucleate pool boiling heat transfer, *J. Heat Transfer* 131 (12) (2009) 121009.
- [14] N. Zuber (Thesis), Hydrodynamic aspects of boiling heat transfer, 1959. No. AECU-4439.
- [15] J.M. Ramilison, P. Sadasivan, J.H. Lienhard, Surface factors influencing burnout on flat heaters, *J. Heat Transfer* 114 (1) (1992) 287–290.
- [16] I. Golobič, K. Ferjančič, The role of enhanced coated surface in pool boiling CHF in FC-72, *Heat Mass Transf.* 36 (6) (2000) 525–531.
- [17] S.W. Ahmad, T.G. Karayiannis, D.B.R. Kenning, A. Luke, Compound effect of EHD and surface roughness in pool boiling and CHF with R-123, *Appl. Therm. Eng.* 31 (11) (2011) 1994–2003.
- [18] C.K. Guan, B. Bon, J. Klausner, R. Mei, Comparison of CHF enhancement on microstructured surfaces with a predictive model, *Heat Transfer Eng.* 35 (5) (2014) 452–460.
- [19] H.S. Ahn, H.J. Jo, S.H. Kang, M.H. Kim, Effect of liquid spreading due to nano/microstructures on the critical heat flux during pool boiling, *Appl. Phys. Lett.* 98 (7) (2011) 071908.
- [20] M.M. Rahman, E. Ölceroğlu, M. McCarthy, Role of wickability on the critical heat flux of structured superhydrophilic surfaces, *Langmuir* 30 (37) (2014) 11225–11234.
- [21] S.J. Kline, F.A. McClintock, Describing uncertainties in single-sample experiments, *Mech. Eng.* 75 (1) (1953) 3–8.
- [22] Y.V. Kalinin, V. Berejnov, R.E. Thorne, Contact line pinning by microfabricated patterns: effects of microscale topography, *Langmuir* 25 (9) (2009) 5391–5397.
- [23] D.L. Cocke, R. Schennach, M.A. Hossain, D.E. Mencer, H. McWhinney, J.R. Parga, M. Kesmea, J.A.G. Gomes, M.Y.A. Mollah, The low-temperature thermal oxidation of copper, Cu_2O , and its influence on past and future studies, *Vacuum* 79 (1) (2005) 71–83.
- [24] S.G. Kandlikar, A theoretical model to predict pool boiling CHF incorporating effects of contact angle and orientation, *J. Heat Transfer* 123 (6) (2001) 1071–1079.

- [25] M.G. Kang, Effect of surface roughness on pool boiling heat transfer, *Int. J. Heat Mass Transfer* 43 (22) (2000) 4073–4085.
- [26] H. Honda, H. Takamastu, J.J. Wei, Enhanced boiling of FC-72 on silicon chips with micro-pin-fins and submicron-scale roughness, *J. Heat Transfer* 124 (2) (2002) 383–390.
- [27] B.S. Kim, H. Lee, S. Shin, G. Choi, H.H. Cho, Interfacial wicking dynamics and its impact on critical heat flux of boiling heat transfer, *Appl. Phys. Lett.* 105 (19) (2014) 191601.
- [28] K. Ferjančič, I. Golobič, Surface effects on pool boiling CHF, *Exp. Thermal Fluid Sci.* 25 (7) (2002) 565–571.
- [29] Y. Haramura, Y. Katto, A new hydrodynamic model of critical heat flux, applicable widely to both pool and forced convection boiling on submerged bodies in saturated liquids, *Int. J. Heat Mass Transfer* 26 (3) (1983) 389–399.
- [30] X. Quan, L. Dong, P. Cheng, A CHF model for saturated pool boiling on a heated surface with micro/nano-scale structures, *Int. J. Heat Mass Transfer* 76 (2014) 452–458.